

Formula sheet CS1520 Statistics

$$\text{Odds Ratio: } \frac{p/1-p}{q/1-q}, \text{ Relative Risk: } \frac{p}{q}$$

$$\text{Mean} = \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i ; \text{ Standard Deviation (SD): } \frac{1}{\sqrt{n-1}} \sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} ;$$

$$\text{Variance} = SD^2 ; \text{ Standard Error For Mean} = \frac{SD}{\sqrt{n}}$$

$$\text{Correlation} = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\frac{1}{\sqrt{n}} \sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} * \frac{1}{\sqrt{n}} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

$$\text{Spearman Correlation} = \text{Correlation}(\text{rank}(X), \text{rank}(Y))$$

Probability Axioms:

Probability of event E: $P(E) \in [0,1]$.

For an event A: $P(\sim A) = 1 - P(A)$ or $P(A^c) = 1 - P(A)$

If $(E_1 \cap E_2) = \emptyset$ (Null event) then: $P(E_1 \cup E_2) = P(E_1) + P(E_2)$

If $(E_1 \cap E_2) \neq \emptyset$ (Null event) then: $P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$

If two events E_1 and E_2 are independent: $P(E_1 \text{ and } E_2) = P(E_1) * P(E_2)$

$$\text{Conditional Probability: } P(A|B) = \frac{P(A \text{ and } B)}{P(B)} = \frac{P(A \cap B)}{P(B)}$$

$$\text{Bayes' Rule: } P(H | E) = \frac{P(H) \cdot P(E|H)}{P(E)}$$

Law of Total Probability: $P(A) = \sum_{j=1}^n P(B_j) \cdot P(A | B_j)$ where $B_1, B_2, \dots, B_n$ form a partition of the sample space.

Distributions

Cumulative Distribution $F(x) = P(X \leq x)$ for a random variable X.

- $F(x) = \int_{-\infty}^x f(t)dt$ for the probability distribution function $f(t)$
- $P(a \leq X \leq b) = F(b) - F(a) = \int_a^b f(x)dx$

$$\text{Expectation } E[X] = \mu_X = \begin{cases} \sum_{x \in S_X} x \cdot f(x) = \sum_{x \in S_X} x \cdot P(X = x) & X \text{ is a discrete RV} \\ \int_{-\infty}^{\infty} x \cdot f(x) dx & X \text{ is a continuous RV} \end{cases}$$

$$\text{Variance } \text{Var}(X) = E[(X - \mu_X)^2]$$

$$\text{Bernoulli Distribution: } E(X) = p, \text{Var}(X) = p(1 - p)$$

Binomial Distribution: $E(X) = np$, $Var(X) = np(1 - p)$

Normal Distribution: $E(X) = \mu$, $Var(X) = \sigma$

Sampling Distribution of Mean under Central Limit Theorem:

$$\lim_{n \rightarrow \infty} \bar{x} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right), \text{ where standard error } SE = \frac{\sigma}{\sqrt{n}}$$

$$\text{Z-score} = Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}, Z \sim N(0, 1),$$

- $z_{\frac{0.1}{2}} = -1.645$ and $z_{1-\frac{0.1}{2}} = 1.645$ where $\alpha = 0.1$
- $z_{\frac{0.05}{2}} = -1.96$ and $z_{1-\frac{0.05}{2}} = 1.96$ where $\alpha = 0.05$
- $z_{\frac{0.01}{2}} = -2.576$ and $z_{1-\frac{0.01}{2}} = 2.576$ where $\alpha = 0.01$

Confidence Interval: $\hat{\theta} \pm K * SE$, for a summary statistic $\hat{\theta}$ and critical value K which depends on the selected α , this defines a $1 - \alpha$ CI.

For a mean estimate $\bar{x}$, we can get the confidence interval, for some α , using $\bar{x} \pm z_{\alpha/2} * \frac{\hat{\sigma}}{\sqrt{n}}$

For a proportion estimate $\hat{p}$, we can get the confidence interval for some α using $\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$